

Janislley Oliveira De Sousa

R&D Tech Leader at Samsung R&D Center| PhD(c) in Electrical Engineering at UFAM

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PROFILE

- R&D Technical Leader at SIDIA - Samsung for industrial and consumer electronics projects in the Latin America market.
- More than 10 years of experience in Embedded electronics development with a demonstrated history of working in the consumer electronics industry. Actuated in more than 80 consumer electronics projects for Samsung.
- He holds a Bachelor's degree in Computer Engineering from UFPB (2017), a Master's degree in Electrical Engineering from UFAM (2023), and is currently pursuing a Ph.D. in Electrical Engineering at UFAM.
- Senior Embedded Electronics Engineer with a sharp interest in Linux systems, especially what is closest to the hardware: Linux kernel, Linux device driver, charging algorithms, and power management systems. Strong expertise in battery and charging technologies (Wired and Wireless Power Systems).
- Worked on the development of over 150 different Samsung consumer electronics projects, from mid-range to flagship models, and contributed to their successful market release for Latin America.
- Authored more than 30 scientific papers in prestigious conferences and journals: https://scholar.google.com/citations?user=Yw4vt_AAAAAJ&hl
- He is an active IEEE Young Professional member with 8 years of volunteer experience, having volunteered for the IEEE Northeast Brazil Section from 2017 to 2020, and has been volunteering with the IEEE Centro-Norte Brasil Section since 2021. He is also a VoLT Graduate (2021) and was the recipient of the IEEE Citizen Empowerment Prize in 2017 and the IEEE MGA Innovation Award in 2020, the IEEE MGA Young Professionals Achievement Award in 2025, and was selected as a 2025 IEEE Young Professional Innovator. As a YP mentor and consultant, he has supported over 400 students and Young Professional members in taking their first steps toward a successful professional career.

KEY SKILLS

- Passionate about writing code, patient, curious, and with a keen eye for detail;
- Able to quickly learn and understand complex technical topics;
- Interested in sciences and technologies, willing to acquire new knowledge;
- Strong interpersonal communication skills (written and verbal);
- Skilled at prioritization and handling of multiple projects simultaneously;
- Creative, analytical abilities, good technical and innovative problem-solving skills.

WORK EXPERIENCE

Company: SIDIA – Samsung
Position: Tech Leader and Senior Embedded Software Engineer
Period: 10/2017 – Current

Main Achievements:

- Lead a team of 14 engineers responsible for porting and bringing up new embedded software requirements for consumer electronics devices for the Samsung project in Latin America.
- Focal point of Development and support of Linux device drivers for battery charging and power management for Samsung devices (Smartphones and Tablets).
- Oversee technical teams and drive the design, development, and optimization of embedded systems.
- Collaborate with cross-functional departments to ensure high-quality, timely product delivery.
- Mentor team engineers, promote best practices, and implement cutting-edge solutions.

Function description:

- Responsible for Linux drivers & Android systems development, utilizing my hardware and software knowledge.
- Development and support of Linux device drivers for battery and charging of Samsung devices (Smartphones and Tablets).
- Debugging and implementation of Charging (SOC PMIC, MST Driver, Wireless Power & USB Charging, IF & MUIC, Fuel Gauge, Battery) and Power consumption-related features for vendors: Qualcomm, LSI, TI, NXP, and Maxim.
- Porting, bringing up, and testing drivers for a new device on short notice and within a tight product release schedule.
- Acted as a Main contact point and trouble-shooter between a specific HW vendor and our SW developer team on the charging part for the Latam Market.
- Support various stages of mobile device development (smartphones, tablets), including preproduction preparation, product launch, and post-release issue tracking and resolution. Possess complete knowledge of the product development life-cycle.
- Worked on the development of over 150 different devices, from mid-range to flagship models, and contributed to their successful market release.
- Outline change requirements to future iterations of the chips in platforms like Mediatek, Qualcomm, and LSI, discuss with vendors and hardware teams, and propose changes in order to prevent future development issues.
- Maintain, improve, and release the common codebase shared across multiple devices.
- Manage overseas teams, guide and assist with new projects, and train teams regarding new concept requirements.
- Offer knowledge and experience-based feedback on requests that deal with various components in a device, from the system to the platform level.

Software / Tools:

- C, C++, Python, Charger, PMIC, Battery, WPC, Fast Charging, Embedded Software, Power Electronics.

Company: Condutor
Position: Jr Software Engineer
Period: 06/2017 – 10/2017

Main Achievements:

- Development of new payment technologies through the internet of things (IoT) using Bluetooth Low Energy Technology.

Function description:

- Developed a new mobile payment system design using an indoor positioning-based service.
- Bluetooth beacon technology was used to determine the current position of a customer checking out. Based on the detected customer position, the system find whether the

customer could legitimately connect with the POS terminal corresponding to the target payment-enabled area through BLE technology.

Software / Tools:

- C, C++, Lua, HTML5, Git, Vim, VS Code.

Company: PETE - PNCA
Position: Jr Software Engineer
Period: 08/2015 – 03/2016

Main Achievements:

- Develop documentation that meets company standards consistently across new products.

Function description:

- During the internship at the PETE company, I performed several activities related to technical documentation for a new robotics kit. Technical guidance services were provided on programming sensors, actuators, and controllers. In addition, training has been developed for customers who bought robotics kits and technology fairs.

Software / Tools:

- C, Git, Vim, UML.

IEEE Volunteering

Local: Centro-Norte Brasil Section Jt Chpt,RA24/CS23/IE13 - Region 9
Position: Treasurer
Period: 09/2025 – Current

- Coordinate project execution for all chapter activities;
- Develop and manage project budgets, track expenditures, and prepare financial reports for leadership review;
- Evaluate funding requests for new and existing projects, balancing resources effectively to maximize impact and align with organizational priorities.

Local: IEEE Young Professionals Representative at Brazil Council - Region 9
Position: Chair
Period: 06/2025 – Current

- Support all 5 YP AG in the Brazil Council;
- Coordinate project execution for all YP projects;
- Oversee the planning and execution of all YP initiatives within the Brazil Council, ensuring alignment with IEEE guidelines and strategic goals;
- Develop and manage project budgets, track expenditures, and prepare financial reports for leadership review;
- Evaluate new and existing projects, balancing resources effectively to maximize impact and align with organizational priorities.

Local: IEEE Young Professionals Centro-Norte Section - Region 9
Position: Treasurer
Period: 01/2025 – Current

- Coordinate project execution for all YP projects at the Centro-Norte Section;
- Oversee the financial planning and execution of all YP initiatives within the Centro-Norte Section, ensuring alignment with IEEE guidelines and strategic goals;

- Develop and manage project budgets, track expenditures, and prepare financial reports for leadership review;
- Evaluate funding requests for new and existing projects, balancing resources effectively to maximize impact and align with organizational priorities.

Local: IEEE Young Professionals Centro-Norte Section - Region 9
Position: Technical Consultant
Period: 01/2021 – Current

- Graduated from the **IEEE VoLT Program** in **2020**.
- Since 2021, he has been actively involved in IEEE Young Professional events at technical conferences, delivering presentations and workshops for young professionals and graduate students. In addition, he has conducted over **20 talks by invitation from conferences and universities** in Brazil and across Latin America. My tasks impacted around **300 young professionals** in he fields of Industrial applications, power electronics, and consumer devices applications. Some examples of talks:
    - https://www.facebook.com/share/p/18St5PWQY6/
    - https://yp.ieee.org/event/coffee-talk-ieee-do-estudante-ao-profissional-contribuicoes-do-ieee-no-meu-caminho-como-engenheiro/
    - https://www.youtube.com/watch?v=qKlS00m-p1Q&pp=ygUJamFuaXNsbGV5
    - https://www.youtube.com/watch?v=texe_Aox4KQ&t=4s&pp=ygUJamFuaXNsbGV5
    - https://www.linkedin.com/posts/janislley_essa-semana-que-passou-ocorreu-o-latinoware-activity-7269041084469555200-46St?utm_source=share&utm_medium=member_desktop
    - https://www.linkedin.com/posts/janislley_pels-ias-ieee-activity-7175581258968621056-e05R?utm_source=share&utm_medium=member_desktop
    - https://www.linkedin.com/posts/janislley_ap%C3%B3s-13-anos-de-concluir-o-curso-t%C3%A9cnico-activity-7174451675305295872-EkAA?utm_source=share&utm_medium=member_desktop
- Participated in the SchnaPYS event with IEEE PELS at COBEP in 2021, 2023, and 2025, meeting with Young Professionals to exchange experiences and innovative ideas for the industry. Also participated in the IEEE CTSoc Young Professional meeting at the 2025 IEEE ICCE conference, where we discussed innovative strategies to promote and develop new engineers in the field of industrial and consumer electronic devices. Selected as a 2025 IEEE Young Professional Innovator for the IEEE VIC summit.

Local: IEEE Young Professionals Nordeste Section - Region 9
Position: Project Coordinator
Period: 02/2018 – 12/2020

- Committed to helping young professionals evaluate their career goals, polish their professional image, and create the building blocks of a lifelong and diverse professional network.
- Contributed to the creation of the following projects: Time4Thesis, CoffIEEEssional, and SolvIEEE it!
- The SolvIEEE it! The project was recognized with the IEEE MGA Innovation Award in 2020.

Local: IEEE Student Branch Federal University of Paraiba (UFPB) - Region 9
Position: Project and Marketing Coordinator
Period: 06/2016 – 01/2018

- Develop students' technical and non-technical skills, promoting interdisciplinary activities to complement, extend, and apply academic knowledge to generate new and better learning opportunities for the entire academic community.

- Received the IEEE Citizen Empowerment Prize for the Public Safety Project at the 1st IEEE Summer School on Smart Cities.

INTERNATIONAL EXPERIENCE

Company: Samsung Electronics
Position: Tech Leader and Senior Embedded Software Engineer
Period: 01/2025 – 01/2025
Local: EUA

Present 3 scientific papers as results of the Samsung innovation projects at the IEEE ICCE 2025.

Company: Samsung Electronics
Position: Embedded Software Engineer
Period: 07/2018 – 08/2018
Local: South Korea

Co-working with Samsung Electronics in South Korea for new mobile projects.
Had the experience of working with other cultural teams from Korea, India, and Vietnam in system development for different Samsung projects.

EDUCATION

Program: PHD(c) of Electrical Engineering
Institution: UFAM - Federal University of Amazonas (2024 - 2027)
Location: Manaus - AM, Brazil

Program: Master of Electrical Engineering
Institution: UFAM - Federal University of Amazonas (2021 - 2023)
Location: Manaus - AM, Brazil

Program: Bachelor's Degree, Computer Engineering
Institution: UFPB - Federal University of Paraíba (2012 - 2017)
Location: João Pessoa - PB, Brazil

Program: High School, Information Technology
Institution: IFPB (2008 - 2011)
Location: Cajazeiras - PB, Brazil

LANGUAGES

English: Professional
Portuguese: Native
Spanish: Intermediate